1. $z = 6 - 1 = 5$

2. $z = 1.5 \cdot 1 - 0.5 = 1$

$\hat{y} = \frac{1}{1+e^{-1}} \approx 0.731$

3. $z = 3 \cdot -2 + 1 = -5$

$a = \max(0, -5) = 0$

4. $L = \frac{1}{2}(1-4)^2 = 4.5$

5. $\frac{\partial L}{\partial \hat{y}} = (\hat{y} - y) = -4$

6. $\frac{\partial L}{\partial w} = \frac{\partial L}{\partial u} \cdot \frac{\partial u}{\partial w} = x(\hat{y} - y)$

$= 2(-3) = -6$

$\hat{y} = 2$

$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial u} \cdot \frac{\partial u}{\partial b} = (\hat{y} - y)$

$= 2 - 5 = -3$

7. $w_{new} = 1 - (0.1 \cdot -6) = 1.6$

$b_{new} = 0 - 0.1 \cdot -3 = 0.3$

8. $\hat{y} = 0 \cdot 1 + 0 = 0$

$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x = (0-2) \cdot 1 = -2$

$\frac{\partial L}{\partial b} = (\hat{y} - y) = (0-2) = -2$

$w_1 \leftarrow 0 - 0.5 \cdot -2 = 1$

$b_1 \leftarrow 0 - 0.5 \cdot -2 = 1$

$\hat{y} = 1 \cdot 1 + 1 = 2$

$\frac{\partial L}{\partial w} = (\hat{y} - y) \cdot x = (2-2) \cdot 1 = 0$

$\frac{\partial L}{\partial b} = (\hat{y} - y) = (2-2) = 0$

$w_2 \leftarrow 1 - 0.5 \cdot 0 = 1$

$b_2 \leftarrow 1 - 0.5 \cdot 0 = 1$